

Stem-Cell Evidence: Benefit, Risk, Uncertainty

LAUNCH • Science • 40 minutes

I can analyse simulated stem-cell evidence by calculating outcomes and identifying benefits, risks and limitations.

Name: _____ Date: _____

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

Word	Meaning
benefit	possible helpful outcome
risk	chance of harm
uncertainty	what is not yet secure

First idea

In a fictional training study, 12 participants improved. Which missing fact is needed first?

- A. How many participants were in the group.
- B. The colour of the data table.
- C. Whether a headline called it a breakthrough.

My first idea and reason:

group	improved	total
treated	12	20
control	6	20

Simulated results table

We do • choose, explain, check

Four of 20 participants experienced the defined adverse event. Which calculation gives the participant percentage?

A. $4 \div 20 \times 100 = 20\%$

B. $20 \div 4 \times 100 = 500\%$

C. 4% because four events occurred

My choice: _____ Evidence or reason:

Benefit-risk-uncertainty analysis lab

Place each evidence statement, then use the frozen set to make one cautious analysis.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
BENEFIT EVIDENCE	
RISK EVIDENCE	
UNCERTAINTY / LIMIT	

Activity cards

1 • Improvement frequency	2 • Adverse-event frequency
12/20 in treatment and 6/20 in comparison improved in the simulation.	4/20 in treatment and 1/20 in comparison had the defined adverse event.
3 • Short follow-up	
Participants were followed for only eight weeks.	

What benefit, risk and uncertainty must appear in a scientifically cautious analysis?

Your lesson record

Analyse the simulated table: calculate four percentages, classify benefit and risk, and state two uncertainties.

Model helps us see

The table makes frequencies, totals and percentages comparable and keeps benefit, risk and uncertainty in the same view.

Model does not show

The numbers are invented for classroom reasoning, the sample is small and no real treatment conclusion can be drawn from them.

Supported

Complete two percentage calculations with the formula shown, then match one benefit, risk and limit.

Numbers are pre-positioned: $\text{frequency} \div 20 \times 100$. Use icons and exact scribing if needed.

Standard

Calculate all four percentages and write a three-part analysis.

Use: The data suggest __. However __. We cannot conclude __ because __.

Stretch

Calculate all outcomes, compare groups and evaluate how sample size and follow-up affect confidence.

Do not add new biology; deepen the evidence judgement and propose one useful next-data need.

Evidence target

Visible percentage working plus a benefit-risk-uncertainty analysis that does not overclaim.

Exit, Audience and evidence • W11L2

Supported: Calculate 6 out of 20 as a percentage.

Standard: State one benefit finding and one uncertainty from the simulated data.

Stretch: Why can the same dataset suggest benefit and still be insufficient for a confident treatment claim?

What I said, and what it changed

Next I will _____.

Adult witness note

What was observed, and with what level of independence?

My evidence and explanation

What does this evidence not show?

Visual reference cards

These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.

group	improved	total
treated	12	20
control	6	20

Totals highlighted

$$12 \div 20 \times 100 = 60\% \cdot 6 \div 20 \times 100 = 30\%$$

improvement recorded more often in the treated group — in this small, short, simulated study

Percentages

Exit • one next step

After the adult has listened, what will you keep, improve or check next?
